

The Fermionic $U(n)^3$ Quiver: BPS Entropy, Concentration, and Chaos

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Abstract. We study a purely fermionic, $\mathcal{N} = 2$ supersymmetric gauged quantum mechanics that, as far as we can tell, has not been considered before: three bifundamental fermions A, B, C of $U(n)^3$ arranged on a triangle quiver, each in p flavours, with the cubic supercharge $Q = \sum C_{abc} \text{Tr}(A^a B^b C^c)$. It is the matrix ($D = 2$) member of the coloured models of Gurau–Witten type, so its large- n limit is planar, and it was chosen because bifundamental matter has no zero weights, which removes the mechanism that makes the BPS window of adjoint matrix models grow with the rank. In the gauge-singlet sector, exact cohomology at $(n, p) = (2, 2)$ and Lanczos diagonalisation at $(2, 3)$ show that all singlet BPS states, 90 and 1680 of them, sit at half filling, and all 90 classes at $(2, 2)$ are fortuitous. At $p = 3$ the singlet multiplet spectrum has Gaussian-orthogonal level statistics over 10 024 levels and small gaps only next to the BPS degree, while $p = 2$ turns out to be special, with a hidden $\mathcal{N} = 4$ supersymmetry on singlets and a spectrum that is not random-matrix like. We then show that the singlet index is exactly the partition function of a three-species log gas on the circle. This proves $I_0(n, 2) = (-1)^n (3n)!/(n!)^3$ and fixes the sign of the index in general, and the large- n saddle gives $\ln |I_0| = F^*(p) n^2 + O(\ln n)$ with $F^* > 0$ exactly when $p > 2$; at $p = 3$, $F^* = 1.2373$, which is 41% of the singlet log-count, and as $p \rightarrow \infty$ the BPS entropy per fermion approaches the $\mathcal{N} = 2$ SYK value $\frac{1}{2} \ln 3$. A parameter-free finite- n evaluation of the same saddle reproduces all 27 non-zero exact indices and singlet counts to within 0.6 in the logarithm. The $p \geq 3$ quiver therefore has macroscopic BPS entropy, concentration and chaos at the sizes we can reach; concentration at $n \geq 3$, fortuity at $p \geq 3$ and genuine near-BPS scaling are the open questions, and we describe how to attack them.

1 Introduction

The fortuity programme proposes that the BPS states responsible for the entropy of supersymmetric black holes are cohomology classes that exist at a given rank but cannot be continued to larger rank, as opposed to monotone classes that are inherited along a tower of ranks [1]. Chang, Chen, Sia and Yang argued that a theory whose BPS states are mostly fortuitous should exhibit R-charge concentration [2]: within each complex, the BPS states sit in a single degree of the R-charge. $\mathcal{N} = 2$ SYK is the paradigm [3]. There the BPS states occupy a window of width $\hat{q}$ around half filling, their number grows exponentially with the number of fermions, and the non-BPS spectrum next to them is chaotic, with the density of states of the $\mathcal{N} = 2$ super-Schwarzian [4]. Chen asked for the simplest *matrix* model with these properties, a “matrix SYK” model: purely fermionic matrix quantum mechanics with a cubic supercharge built from traces [5]. He showed that a single adjoint fermion already has fortuitous BPS states, although they spread over $N + 1$ charges, and suggested that three adjoint flavours might concentrate.

Our earlier reports [6, 7] analysed this adjoint family in detail and reached a negative conclusion. At the maximal weight the BPS window has width $W = r(w_s - 1) + 1$ for a gauge algebra of rank r , where the width $w_s \geq 2$ contributed by a single Cartan site is controlled by one bilinear form. The window therefore grows with the rank, and a concentrated large- N limit is impossible for adjoint matter at fixed degree of the supercharge. The mechanism lives on the Cartan: the adjoint representation has zero weights, and at the maximal weight the complex is a product of Cartan sites, each of which contributes its own spread in charge. This points to an obvious way out, namely to build the cubic supercharge from matter without zero weights. The simplest such matter is bifundamental, and the simplest cubic gauge invariant of bifundamentals is the trace around a triangle, $\text{Tr}(ABC)$.

This report collects what we have learned about the resulting model. We describe it in Section 2 and place it in the literature in Section 3. Section 4 presents the exact and numerical results on BPS states at small rank: singlet indices, concentration at $(n, p) = (2, 2)$ and $(2, 3)$, and a fortuity test. Section 5 turns to the non-BPS singlet spectrum, where $p = 3$ is chaotic and $p = 2$ has a hidden extended supersymmetry. Section 6 contains the main analytic result: the singlet index is the partition function of a three-species log gas, which we solve at large n . Section 7 assesses the prospects of the $p \geq 3$ model and lists the next steps.

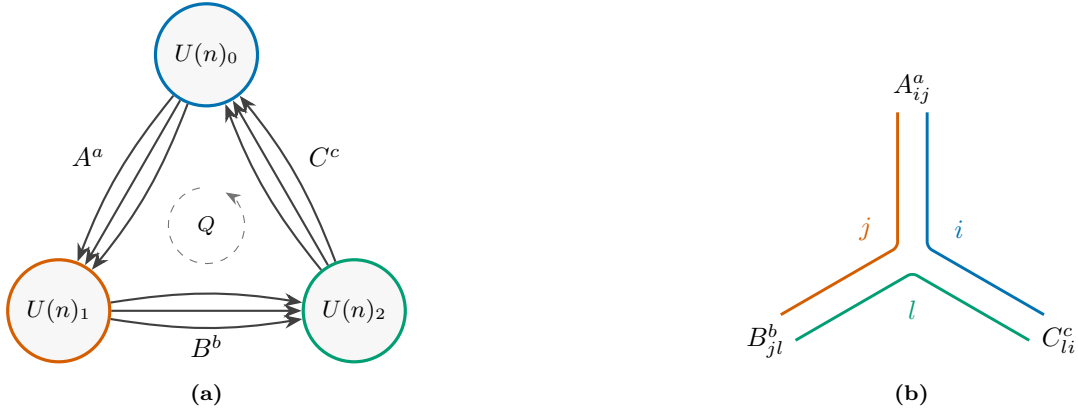


Figure 1: (a) The quiver. An arrow from node v to node $v + 1 \pmod{3}$ is a bundle of p fermionic bifundamentals in $(\mathbf{n}_v, \bar{\mathbf{n}}_{v+1})$, drawn here for $p = 3$. The supercharge (2) is the gauge-invariant loop around the triangle, with a general flavour tensor C_{abc} . (b) The interaction as a ribbon vertex. Each field carries the indices of the two nodes it joins, and each index line closes around the vertex. This is the $D = 2$ member of the coloured models of [8]; at large n only planar ribbon graphs survive.

2 The model

The gauge group is $U(n)_0 \times U(n)_1 \times U(n)_2$, with fundamental indices i, j and l for the three nodes. The matter consists of complex fermions

$$A^a_{ij} \in (\mathbf{n}, \bar{\mathbf{n}}, \mathbf{1}), \quad B^a_{jl} \in (\mathbf{1}, \mathbf{n}, \bar{\mathbf{n}}), \quad C^a_{li} \in (\bar{\mathbf{n}}, \mathbf{1}, \mathbf{n}), \quad a = 1, \dots, p, \quad (1)$$

transforming as $A \rightarrow g_0 A g_1^{-1}$, $B \rightarrow g_1 B g_2^{-1}$ and $C \rightarrow g_2 C g_0^{-1}$, with canonical anticommutators such as $\{A^a_{ij}, (A^b_{i'j'})^\dagger\} = \delta^{ab} \delta_{ii'} \delta_{jj'}$. We regard A, B and C as creation operators acting on a Fock vacuum annihilated by their adjoints. The Hilbert space is the exterior algebra $\mathcal{H} = \Lambda^\bullet V$ on the $3pn^2$ modes of $V = \mathbb{C}^p \otimes [(\mathbf{n}, \bar{\mathbf{n}}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{n}, \bar{\mathbf{n}}) \oplus (\bar{\mathbf{n}}, \mathbf{1}, \mathbf{n})]$, of dimension 2^{3pn^2} . The dynamics is fixed by a single supercharge,

$$Q = \sum_{a,b,c=1}^p C_{abc} \text{Tr}(A^a B^b C^c) = \sum_{a,b,c} C_{abc} A^a_{ij} B^b_{jl} C^c_{li}, \quad H = \{Q, Q^\dagger\} \geq 0. \quad (2)$$

Because Q is a product of three creation operators it acts on $\mathcal{H}$ as exterior multiplication by a fixed gauge-invariant three-form, and $Q^2 = 0$ holds identically: no condition is imposed on the couplings, which form an arbitrary tensor $C \in \mathbb{C}^p \otimes \mathbb{C}^p \otimes \mathbb{C}^p$. (The three fields are distinct, so no symmetrisation is implied, and flavour rotations $U(p)^3$ act on C .) All numerical results below use generic couplings, mostly random integers between 1 and 5 with a fixed seed, and several Gaussian real and complex sets as checks. Figure 1 shows the quiver and the interaction vertex.

The theory has a $U(1)^3$ of fermion numbers N_A, N_B, N_C , each raised by one unit by Q . Two combinations of them, $N_A - N_C$ and $N_B - N_A$, are the central charges of the gauge nodes, so gauge singlets have $N_A = N_B = N_C = m$ and total fermion number $k = N_A + N_B + N_C = 3m$. The total fermion number plays the role of the R-charge, and Q raises it by $\hat{q} = 3$. BPS states, $H\psi = 0$, are in one-to-one correspondence with Q -cohomology classes, and we write h^k for their number in degree k and $n(k)$ for the number of gauge singlets in degree k . The pairing with the top form of $\Lambda^\bullet V$ exchanges k and $3pn^2 - k$, so $n(k)$ is palindromic and $h^k = h^{3pn^2 - k}$, and half filling is $k_* = 3pn^2/2$. Non-BPS states come in Q -pairs $(\psi, Q\psi)$ in degrees k and $k+3$ with a common energy, and we label the pair by its charge relative to half filling, $q = k + \frac{3}{2} - \frac{3}{2}pn^2$, following [4]. Since $(-1)^k$ anticommutes with Q , the singlet index

$$I_0(n, p) = \sum_m (-1)^m n(3m) = \sum_m (-1)^m h_{\text{sing}}^{3m} \quad (3)$$

is a lower bound on the number of singlet BPS states. The bound is saturated when the singlet BPS states sit in a single degree, which is what concentration means here. In $\mathcal{N} = 2$ SYK with $\hat{q} = 3$ the BPS states fill three adjacent charges in the ratio 1:2:1 [3, 4]; gauge singlets live only in degrees $k \equiv 0 \pmod{3}$, so for even pn^2 that window contains exactly one singlet degree, half filling, and concentration of the singlet sector means that every

singlet BPS state has $k = 3pn^2/2$. By Molien–Weyl, $n(t) = \sum_k n(k)t^k$ is the unitary matrix integral

$$n(t) = \int \prod_{v=0}^2 dU_v \prod_{v=0}^2 \det(1 + t U_v \otimes U_{v+1}^\dagger)^p, \quad I_0 = n(-1), \quad (4)$$

over Haar measure, with v taken mod 3. This integral is the starting point of Section 6.

Two structural remarks explain why the model was chosen and where its interesting range of parameters lies. The weights of the bifundamental modes are differences $e_i - f_j$ of weights of two different nodes and never vanish, so the zero-weight (Cartan) modes on which the window-widening mechanism of the adjoint models lives are simply absent. This removes the obstruction of [6, 7]; it does not by itself imply concentration. Second, the gauge group has dimension $3n^2$ against $3pn^2$ fermion modes, a ratio of p rather than a large number, so the singlet constraint is severe. At $p = 1$ it leaves essentially nothing; at $n = 2$ and 3 every irrep’s complex has index ± 1 , the singlet index vanishes, and concentration is vacuous. The model becomes interesting at $p \geq 2$.

3 Relation to previous work

Witten obtained the large- N physics of SYK without disorder from a coloured model of $q = D+1$ fermion fields of rank D , every pair of which shares one index [8]. For $q = 3$ and $D = 2$ the three fields are exactly bifundamentals of a three-node quiver and the vertex is $\text{Tr}(ABC)$. Witten’s melonic dominance argument, however, requires $D \geq 3$ (his eq. (3.10)); at $D = 2$ the relevant combinatorial invariant reduces to the genus of ribbon graphs and the large- n limit is planar. Our model is the complex, flavoured, $\mathcal{N} = 2$ supersymmetric version of this matrix member of the coloured family. Supersymmetric tensor versions of SYK exist with $\mathcal{N} = 1$ supersymmetry and higher-rank tensors [9]. Chang, Colin-Ellerin and Rangamani showed that $\mathcal{N} = 2$ supertensor models with dynamical bosons break supersymmetry at large N , and noted that the purely fermionic $\mathcal{N} = 2$ model with an odd-degree supercharge has no evident melonic tensor version (their eq. (6.4)) [10]. Biggs, Lin and Maldacena have since closed that gap melodically, replacing tensor contractions by $SU(2)$ $3j$ symbols [11]. The quiver is a planar counterpart: purely fermionic, cubic and $\mathcal{N} = 2$, without dynamical bosons and hence without the supersymmetry-breaking mechanism of [10].

The same three-node quiver with a closed loop and a cyclic cubic superpotential is famous in a different guise, as the quiver quantum mechanics of three-centre BPS bound states in string theory. For bosonic chiral multiplets on abelian nodes, Bena, Berkooz, de Boer, El-Showk and Van den Bleeken found exponentially many “pure-Higgs” states that are invisible on the Coulomb branch and are natural candidates for single-centre black hole microstates [12]; they also remarked that their counting formula suggests an origin in fermionic degrees of freedom. Our model is a different theory, purely fermionic and non-abelian, whose BPS states are the cohomology of a finite Fock space with no Higgs or Coulomb branch, and we claim no equivalence; a naive comparison of our $n = 1$ counts with theirs already fails.

On the fortuity side, the monotone/fortuitous dichotomy and the covering construction are due to Chang and Lin [1], and concentration as its signature in SYK to [2]. Chen constructed fortuitous states in a single adjoint matrix and used a unitary matrix integral to follow them as the rank changes [5]. Tierz computed BPS spectra of $\text{Tr}\Psi^p$ models and formulated fortuity through projections between ranks, which are cochain maps [13]; we use his definition. Choi, Choi and Kim studied fortuity in the bifundamental conifold theory and introduced “generalised monotones” dressed by baryons [14], which bears directly on our fortuity test (Section 4.3). The natural tool for large n is the quantum mechanical matrix bootstrap [15, 16, 17], which Chen already suggested for matrix SYK models; our sector-resolved version is described in [7]. A literature search in October 2026 found no previous study of the fermionic quiver itself, although absence from searches is not proof.

4 BPS states at small rank

4.1 Singlet indices

The singlet multiplicities $n(k)$ follow from the dual Cauchy identity, $\Lambda(\mathbb{C}^p \otimes V_v \otimes \bar{V}_{v+1}) = \bigotimes_{f=1}^p \bigoplus_\lambda S_\lambda V_v \otimes S_{\lambda'} \bar{V}_{v+1}$, which reduces the singlet count to products of Littlewood–Richardson coefficients. We compute these by exact quadrature on the torus and assemble the index with a transfer matrix over $U(n)$ irreps that adds one flavour at a time,

$$I_0 = \text{Tr}[(R^{(p)})^3], \quad R^{(q)} = \sum_\lambda (-1)^{|\lambda|} L_\lambda^T R^{(q-1)} L_{\lambda^T}, \quad (L_\lambda)_{\mu\nu} = c_{\mu\lambda}^\nu, \quad (5)$$

(n, p)	modes $3pn^2$	singlet states	$I_0(n, p)$	$\ln I_0 $ per mode	$(-1)^{pn^2/2}$
(1, 2)	6	10	-6	0.299	-
(2, 2)	24	1 274	90	0.187	+
(3, 2)	54	2 115 632	-1 680	0.138	-
(4, 2)	96	5.59×10^{10}	34 650	0.109	+
(1, 3)	9	56	0	-	$(pn^2 \text{ odd})$
(2, 3)	36	889 040	1 680	0.206	+
(3, 3)	81	4.90×10^{12}	0	-	$(pn^2 \text{ odd})$
(4, 3)	144	1.08×10^{22}	9 434 197 872	0.159	+
(1, 4)	12	346	90	0.375	+
(2, 4)	48	9.70×10^8	519 750	0.274	+
(3, 4)	108	2.99×10^{19}	199 090 719 360	0.241	+
(1, 5)	15	2 252	0	-	$(pn^2 \text{ odd})$
(2, 5)	60	1.37×10^{12}	95 866 056	0.306	+
(1, 6)	18	15 184	-1 680	0.413	-
(2, 6)	72	2.29×10^{15}	28 686 212 100	0.334	+

Table 1: Exact gauge-singlet indices and singlet counts. The $n = 1$ rows are Dixon’s sum $\sum_m (-1)^m \binom{p}{m}^3$ and the Franel numbers $\sum_m \binom{p}{m}^3$. At $p = 1$ the index vanishes for $n = 2, 3, 4$. The last column is the sign that a singlet BPS sector concentrated at half filling would give; it agrees with every non-zero entry.

in exact integer arithmetic. This reaches $(n, p) = (4, 3)$ and $(3, 4)$ and reproduces every value obtained earlier by direct weight counting. Table 1 lists the results together with the total singlet counts.

Three features of the table are worth stating at once. First, the $p = 2$ entries are $(-1)^n (3n)!/(n!)^3$, a formula that we found numerically and prove in Section 6. The first few indices at $p = 3$ coincided with this pattern, $|I_0(2, 3)| = |I_0(3, 2)| = 1680$, which briefly suggested that the index depends only on np and grows only exponentially in n ; the value at $(4, 3)$, which is about 550 times larger than the $p = 2$ -type value, shows that the coincidence was accidental. Second, whenever pn^2 is even the sign of I_0 is $(-1)^{pn^2/2}$, which is what a singlet BPS sector concentrated at half filling would produce, since half filling has $m = pn^2/2$; for odd pn^2 the index vanishes identically, because the particle-hole map $m \rightarrow pn^2 - m$ pairs degrees of opposite sign. Third, at fixed n the index per mode rises with p towards the singlet entropy per mode, which already suggests that large p is the SYK-like direction.

4.2 Concentration at $(2, 2)$ and $(2, 3)$

At $(n, p) = (2, 2)$ the singlet cohomology can be computed with exact (modular) ranks. There are exactly 90 singlet BPS states, all at $k = 12$, which is half filling, so the index is saturated. At $(2, 3)$ the Fock space has 2^{36} states and the singlet sector at half filling has 300 584, which calls for a different method. We restrict to the zero-weight subspace of each degree, which contains every irrep, and to the vectors invariant under the Weyl group $(S_2)^3$, which commutes with Q and fixes singlets; this reduces the dimension by about a factor of eight. On these vectors the operator

$$P = \sum_{v=0}^2 \|E_{12}^{(v)} \psi\|^2 = \sum_v \langle \psi | J_v^2 | \psi \rangle \quad (6)$$

is the sum of the $SU(2)$ Casimirs of the three nodes, because $J_- J_+ = J^2$ at $J_z = 0$. It commutes with H , vanishes exactly on singlets and is at least 2 on every non-singlet. The kernel of $K = H + \mu P$ is therefore the singlet BPS space, and every eigenvalue of K below 2μ is an exact singlet energy. Lanczos diagonalisation of K in the degrees $k = 9, 12$ and 15 (Weyl-symmetric dimensions 5.6×10^4 , 5.1×10^5 and 1.8×10^6), and dense diagonalisation for $k \leq 6$, finds no zero modes, with lowest eigenvalues 96.5, 0.3348 and 0.05386 at $k = 9, 12$ and 15 ; reruns at $k = 12$ and 15 with different starting vectors reproduce them to nine digits. Particle-hole symmetry excludes BPS states at $k \geq 21$ as well, so all singlet BPS states sit at $k = 18$, and by the index there are exactly $h_{\text{sing}}^{18} = |I_0| = 1680$ of them. Figure 2 summarises both cases: the singlet sector spreads over 13 degrees at $(2, 3)$, but its BPS states occupy one.

A direct test at $n = 3$ is out of reach for now. At $(3, 2)$ the zero-weight sectors near half filling hold 5×10^{10} states, although the singlet subspaces are only about 5×10^5 ; what is missing is a cheap explicit basis of singlets.

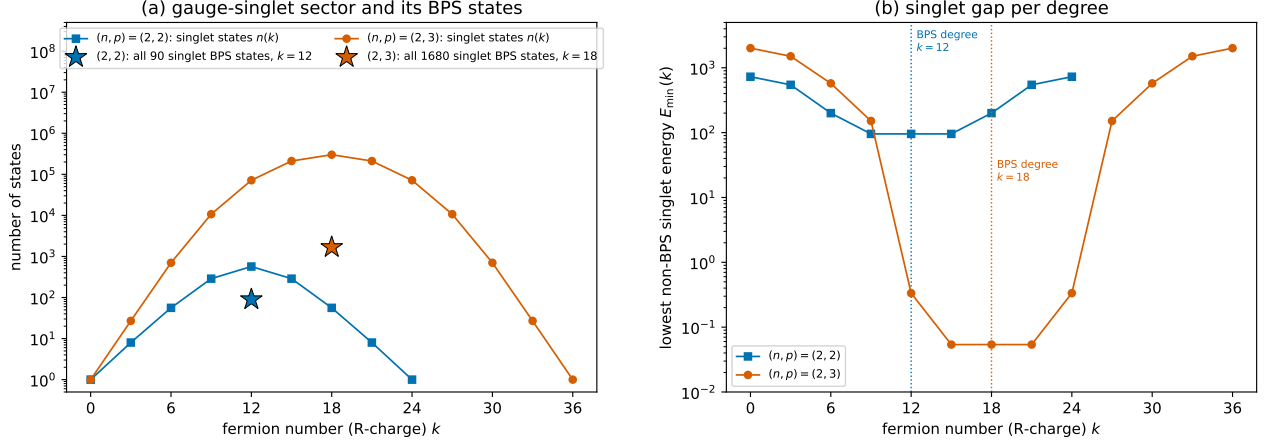


Figure 2: Concentration of the singlet BPS states at $n = 2$. (a) The number of gauge singlets in each degree k (lines) and the singlet BPS states (stars): 90 at $k = 12$ for $p = 2$ (exact ranks) and 1680 at $k = 18$ for $p = 3$ (Lanczos); there are none in any other degree. (b) The lowest non-BPS singlet energy in each degree. At $p = 3$ the gap collapses by four orders of magnitude next to the BPS degree, to 0.054 at $k = 15, 18, 21$ and 0.335 at $k = 12, 24$, while at $p = 2$ it never falls below 95.6.

The index there, $I_0(3, 2) = -1680$, has the sign that concentration at half filling ($k = 27$, $m = 9$) would give, which is consistent with concentration but not evidence for it.

4.3 Fortuity at $(2, 2)$

Following Tierz [13], we call a rank-2 class monotone only if it lies in the image of the projection $\pi_{3 \rightarrow 2}$ that deletes every monomial involving a third index on any node. Because Q contains only creation operators, $\pi Q^{(3)} = Q^{(2)}\pi$, so the projection is a cochain map. Writing $Q^{(3)} = Q^{(2)} + Q_{\text{new}}$, a closed rank-2 state z_0 is the projection of a closed rank-3 state if and only if $Q_{\text{new}}z_0$ is exact in the subspace of states that carry index-3 modes. Grading that subspace by the number of index-3 modes, the first condition arises with two such modes, and for couplings of full rank in the summed flavour index it reduces to the vanishing of the BPS three-point function $\langle \beta | X | z_0 \rangle$ for every single letter X and every BPS state β in degree $k + 1$. A non-vanishing value obstructs the lift. By gauge covariance it suffices to test one component of each letter. At $(2, 2)$, each of the six letters obstructs a 54-dimensional subspace of the 90 classes, different letters obstruct different subspaces, and together they obstruct all 90. Since the obstruction appears already at the first order of the lift, none of the 90 classes lies in the image of $\pi_{3 \rightarrow 2}$: every singlet BPS class at $(2, 2)$ is fortuitous.

This result is expected rather than discriminating, and the reason can be made precise. The projections preserve the degree, while the half-filling degree $k_* = 3pn^2/2$ moves by $3p(2n+1)/2$ when the rank grows by one, more than the width of any concentrated window. A rank- n class of degree $k_*(n)$ can only be monotone if there is a class of the same degree at rank $n+1$, so concentration at rank $n+1$ implies that every singlet BPS class at rank n is fortuitous; in this sense fortuity follows from concentration, much as in Chang–Chen–Sia–Yang’s argument for generic SYK [2]. At $(2, 2) \rightarrow (3, 2)$ the relevant rank-3 singlet space, at $k = 12$, has only 3494 states, so the statement can be checked directly. The $(2, 2)$ test itself uses one coupling and floating-point ranks, and it is in the ordinary sense. Under the refined criterion of [14], classes of the form (determinant ground state) $\times$ (n -independent excitation) would count as generalised monotones; with fermionic letters many determinants vanish, and whether any of the 90 classes is of that form has not been analysed.

5 The singlet spectrum: chaos at $p = 3$, hidden supersymmetry at $p = 2$

The exactness of the Casimir penalty makes the full singlet spectrum accessible at $n = 2$. Where the singlet subspace is small we construct it explicitly as $\ker P$ and diagonalise $(QU_k)^T(QU_k)$ on a singlet basis U_k , whose non-zero eigenvalues are the multiplet energies of the pair $(k, k+3)$. For the largest pair at $(2, 3)$, $(9, 12)$ with 10700 singlets in the lower degree, we project random vectors onto singlets with the exact polynomial

$\prod_v \prod_{j=1}^6 (1 - J_v^2/j(j+1))$, which leaves a non-singlet residual of 1.4×10^{-15} , and compress $Q^T Q$ onto their span. Every count checks: the rank of the projected vectors equals $n(9)$, the kernel equals the rank of the previous map, and the lowest level, 151.34, agrees with Lanczos.

At $p = 3$ the multiplet spectra are chaotic. The mean ratio of consecutive level spacings over the central 80% of the 10 024 multiplets of the pair (9, 12) is $\langle r \rangle = 0.5268 \pm 0.0028$. The Gaussian-orthogonal value for matrices of the same size is 0.5343 ± 0.0026 and the Poisson value 0.3869 ± 0.0042 , so the spectrum is about 30 standard deviations from Poisson and within two of the orthogonal class. The smaller pair (6, 9), with 676 levels, gives 0.507 ± 0.011 , and the locally unfolded spacing distribution follows the Wigner surmise (Figure 3c). The orthogonal class is the expected one, since for real couplings H is a real matrix in the occupation-number basis.

The near-BPS structure is equally clear. The lowest multiplet energies of the pairs at (2, 3) are 0.05386 for $|q| = 1.5$, the pairs adjacent to the BPS degree, then 0.3348 for $|q| = 4.5$, and then a jump to the microscopic scale set by the couplings: 151.34, 575.23, 1506 and 2008 for $|q| = 7.5, 10.5, 13.5$ and 16.5. Small gaps exist, and only next to the BPS degree (Figure 2b). A quantitative test of the Turiaci–Witten density [4], $\rho_q(E) \propto \sinh(2\pi\sqrt{(E - E_0(q))/E_s})/E$ with $E_0(q) = q^2 E_s/4\hat{q}^2$, does not succeed at this size. Fixing $E_s = 0.862$ from the $|q| = 1.5$ edge predicts $E_0 = 0.485$ at $|q| = 4.5$, against 0.335 measured, an edge ratio of 6.2 instead of 9; fixing the normalisation from the eleven lowest $|q| = 1.5$ levels predicts 5.4, 258, 2000 and 8.7×10^4 multiplets with $|q| = 4.5$ below $E = 1.0, 2.17, 3.11$ and 5.38, against 2, 3, 5 and 7 measured (Figure 3d). This is not surprising. The Turiaci–Witten density is a statement at large e^{S_0} in which neighbouring charge sectors carry the same entropy, whereas at (2, 3) the singlet sectors at $k = 12, 15$ and 18 hold 71 685, 211 113 and 300 584 states, changing by factors of three to four per step. The test needs larger n .

The case $p = 2$ behaves very differently, and the reason is an unexpected symmetry. The (2, 2) spectra are full of exact coincidences between different pairs, and a search over cubic supercharges $\tilde{Q} = Q(\tilde{C})$ that commute with H on singlets finds exactly one besides Q , namely $\tilde{C} = (\varepsilon \otimes \varepsilon \otimes \varepsilon)\tilde{C}$ with ε the 2×2 antisymmetric tensor. On every singlet sector, to 10^{-15} , and for integer, Gaussian real and Gaussian complex couplings,

$$\tilde{Q}^2 = 0, \quad \{Q, \tilde{Q}\} = 0, \quad \{Q^\dagger, \tilde{Q}\} = 0, \quad \{\tilde{Q}, \tilde{Q}^\dagger\} = H, \quad (7)$$

which is an $\mathcal{N} = 4$ algebra. Off the singlets the last two relations fail by about 30% of $|H|$, so the extended supersymmetry is a property of the gauged model, closing only up to gauge transformations. The singlet spectrum organises completely into long $\mathcal{N} = 4$ multiplets spanning degrees $k, k + 3, k + 3$ and $k + 6$, with $n(k) = b(k) + 2b(k - 3) + b(k - 6) + h(k)$ in every degree. Once the flavour symmetry $F = \varepsilon$ on every edge (or the corresponding antiunitary symmetry for complex couplings) is resolved, the level statistics are still not random-matrix like in any symmetry class, with coupling-independent exact degeneracies, and within the span of all gauge-invariant operators with at most four fermions the commutant of H on the $k = 9$ singlets is just $\{1, H\}$. At $p = 3$ the same search finds only Q . The hidden supersymmetry is plausibly tied to the pseudoreality of the $SU(2)$ flavour doublet, but we have not derived it, and we have checked it only at $n = 2$.

6 Large n : the singlet index as a three-species log gas

6.1 An exact rewriting

Write $x_{vi} = e^{i\theta_{vi}}$ for the eigenvalues of U_v in (4) at $t = -1$. Weyl's integration formula introduces $|\Delta(x_v)|^2/n!$ for each node, and the edge factors become $\prod_{ij} (1 - e^{i(\theta_{vi} - \theta_{v+1,j})})^p$. For real φ , $1 - e^{i\varphi} = 2 \sin(\varphi/2) e^{i(\varphi - \pi)/2}$, and the φ -dependent phases cancel around the triangle, since each node is the source of one edge and the target of another. Using $|x_a - x_b| = |2 \sin \frac{\theta_a - \theta_b}{2}|$,

$$I_0(n, p) = \frac{(-i)^{3pn^2}}{(n!)^3} \int \prod_{v,i} \frac{d\theta_{vi}}{2\pi} e^{L(\theta)} \sigma(\theta)^p, \quad (8)$$

$$L = \sum_{\substack{\text{pairs at} \\ \text{the same node}}} 2 \ln |2 \sin \frac{\theta_a - \theta_b}{2}| + p \sum_{\substack{\text{pairs at} \\ \text{different nodes}}} \ln |2 \sin \frac{\theta_a - \theta_b}{2}|,$$

with $\sigma = \prod_v \prod_{ij} \text{sgn} \sin \frac{\theta_{vi} - \theta_{v+1,j}}{2}$. This is the partition function of a gas of $3n$ charges of three species on the circle: charges of the same species repel with weight 2 and charges of different species with weight p , and for odd p there is an additional sign. Because every pair of distinct nodes is joined by exactly one edge, e^L is symmetric under permutations of the species, and only σ remembers the orientation of the quiver. The singlet count is the same integral with $|2 \cos \frac{\theta_a - \theta_b}{2}|$ for pairs at different nodes, which attract, and no phase.

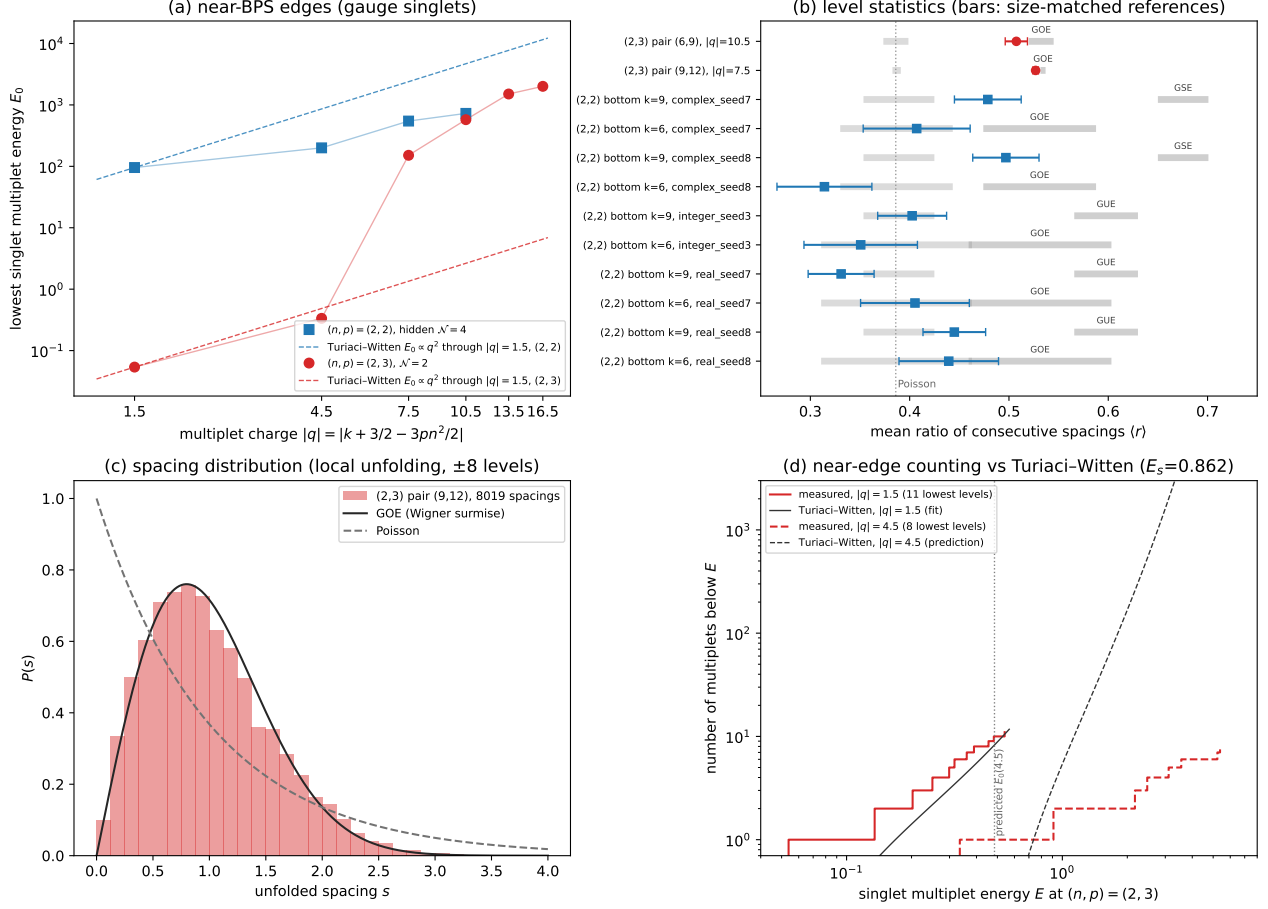


Figure 3: Singlet spectra at $n = 2$. (a) Lowest singlet multiplet energy of each Q -pair against its charge $|q|$, with the Turiaci–Witten q^2 law drawn through the $|q| = 1.5$ point. (b) Mean spacing ratios of every symmetry-resolved ensemble, with size-matched random-matrix and Poisson references. (c) Spacing distribution of the 10024 multiplets of the pair (9, 12) at (2, 3), locally unfolded, against the Wigner surmise and Poisson. (d) Counting functions of the two innermost multiplet sectors at (2, 3) against the Turiaci–Witten density, normalised on $|q| = 1.5$; the $|q| = 4.5$ curve is a prediction.

Several exact statements follow immediately. At $p = 2$ every pair carries weight 2 and $\sigma^2 = 1$, so e^L is the Vandermonde $|\Delta_{3n}(x)|^2$ of all $3n$ eigenvalues together. Its integral is $(3n)!$, and

$$I_0(n, 2) = (-1)^n \frac{(3n)!}{(n!)^3} \quad \text{for every } n. \quad (9)$$

As measured by the index, the singlet BPS entropy of the $p = 2$ model is therefore $3n \ln 3 + O(\ln n)$, which is not macroscopic. The same manipulation, using $(1 - x\bar{y})^2 = -x\bar{y}|x - y|^2$ on the unit circle, gives for general p

$$I_0(n, p) = (-1)^n \frac{(3n)!}{(n!)^3} \mathbb{E}_{\text{CUE}(3n)} \left[\prod_v \prod_{i,j} (1 - x_{vi} \bar{x}_{v+1,j})^{p-2} \right], \quad (10)$$

where the $3n$ eigenvalues of a Haar-random $U(3n)$ matrix are split into three labelled groups. For even p the integrand of (8) is positive, so $I_0 \neq 0$ with sign $(-i)^{3pn^2} = (-1)^{pn^2/2}$; for odd p and odd n the prefactor is $\pm i$ while the integral is real, so $I_0 = 0$; and for odd p , Cauchy–Schwarz in p gives $|I_0(n, p)|^2 \leq |I_0(n, p-1)| |I_0(n, p+1)|$, which holds in the data with ratios 16.6 at (2, 3) and 1.62 at (2, 5). These are exactly the sign rules observed in Table 1.

p	F^* (index)	arc half-width	G^* (singlets)	F^*/G^*	$F^*/3p$	$G^*/3p$
2	0 (exact)	$\pi/3$	1.4244	0	0	0.237
3	1.2373	1.0276	3.0427	0.41	0.137	0.338
4	2.5550	0.9717	4.7688	0.54	0.213	0.397
5	3.9297	0.9132	6.5623	0.60	0.262	0.437
6	5.3455	0.8606	8.4015	0.64	0.297	0.467
8	8.2619	0.7747	12.1714	0.68	0.344	0.507
16	20.4849	0.5796	27.8331	0.74	0.427	0.580
∞	$\frac{3p}{2} \ln 3 - \frac{3}{2} \ln \frac{2p}{3} - \frac{9}{4}$	$\sqrt{6/p}$	$3p \ln 2 - \frac{3}{2} \ln \frac{p}{2} - \frac{9}{4}$	$\frac{\ln 3}{2 \ln 2}$	$\frac{1}{2} \ln 3$	$\ln 2$

Table 2: Leading large- n coefficients, $\ln |I_0| = F^* n^2$ and $\ln(\#\text{singlets}) = G^* n^2$ up to $o(n^2)$, from the continuum solution, with the large- p asymptotics in the last row. The last two columns are per fermion mode.

6.2 The saddle point

At large n , with normalised eigenvalue densities ρ_v , $L = n^2 F[\rho] + O(n \ln n)$ with

$$F[\rho] = \sum_v \langle \rho_v, f \rho_v \rangle + p \sum_{v < w} \langle \rho_v, f \rho_w \rangle, \quad f(\alpha) = \ln \left| 2 \sin \frac{\alpha}{2} \right| = - \sum_{m \geq 1} \frac{\cos m\alpha}{m}, \quad (11)$$

and since $(n!)^3 = e^{O(n \ln n)}$, $\ln |I_0| = n^2 \max F + o(n^2)$ for even p by standard log-gas large deviations. In terms of the moments $u_{v,m} = \int \rho_v e^{im\theta}$, $F = - \sum_m \frac{1}{m} u_m^\dagger [\mathbb{K} + \frac{p}{2}(J - \mathbb{K})] u_m$ with J the all-ones 3×3 matrix, whose eigenvalues are $1+p$ for the species-symmetric combination and $1-p/2$ twice. For $p < 2$ the uniform distribution is a stable maximum with $F = 0$; at $p = 2$ it is marginal; for $p > 2$ it is unstable towards configurations in which the species separate, in the spirit of the Hagedorn instability of weakly coupled gauge theories [18]. The maximum then sits on the boundary of the positivity constraint: the eigenvalues of the three nodes occupy three disjoint arcs rotated by $2\pi/3$ with respect to each other (Figure 4d). Physically the holonomies sit near $U_v \approx e^{2\pi i v/3}$, which turns the factor (-1) that each bifundamental fermion carries in the index into a phase $e^{\mp 2\pi i/3}$ and so evades the cancellations between even and odd states that keep the index small at $p \leq 2$.

We solve the continuum problem by expanding the density of one arc in Chebyshev polynomials. The Cauchy part of the Euler–Lagrange equation is treated exactly and the smooth part by quadrature; the arc width follows from the condition that the edges be soft, and F^* is three times the Lagrange multiplier. The solution converges to twelve digits, and the effective potential is flat on the support to 10^{-15} and lower in the gaps. Independently, Newton maximisation of the discrete L up to $n = 2048$ extrapolates to the same coefficients to 10^{-9} or better, and for $n \leq 12$ no unconstrained random start finds a higher maximum than the symmetric one. The singlet count is treated in the same way: there the three species coincide on a single arc. Table 2 gives the coefficients.

Both limits can be derived analytically and are reproduced by the numerics. Near $p = 2$ the uniform arcs of length $2\pi/3$ give the bound $F^*(p) \geq (p/2 - 1) 39\zeta(3)/2\pi^2$, which is tight as $p \rightarrow 2^+$. At large p the arcs shrink to width $O(p^{-1/2})$, each becomes a semicircle of radius $\sqrt{6/p}$ in the effective harmonic potential of the other two, and $F^* = \frac{3p}{2} \ln 3 - \frac{3}{2} \ln \frac{2p}{3} - \frac{9}{4} + O(1/p)$; the singlet count behaves in the same way with $\ln 2$ in place of $\frac{1}{2} \ln 3$. The leading terms have a simple meaning. $3p \ln 2 n^2$ is the logarithm of the dimension of the whole Fock space, and $\frac{3p}{2} \ln 3 n^2$ is what $3pn^2$ fermion modes contribute when each carries the weight $|1 - e^{-2\pi i/3}| = \sqrt{3}$. The BPS entropy per fermion therefore approaches $\ln(2 \cos \frac{\pi}{6}) = \frac{1}{2} \ln 3$, which is precisely the ground-state entropy per complex fermion of $\mathcal{N} = 2$ SYK with $\hat{q} = 3$ [3], and the ratio F^*/G^* approaches the SYK ratio $\ln \sqrt{3}/\ln 2 = 0.79$. The same constant appears in Tierz’s large- N result for the irrep-summed indices of the single-matrix $\text{Tr} \Psi^3$ model [13]. At finite p the gauge constraint costs $\frac{1}{2p} \ln \frac{2p}{3} + \frac{3}{4p}$ per mode, which is large at $p = 3$: there the singlet BPS entropy per mode is 0.137, a quarter of the SYK value, but it is a finite fraction, 41%, of the singlet log-count.

6.3 Finite n and the test at $(4, 3)$

The coefficient F^* alone cannot be compared with the data at $n \leq 4$: at $(4, 3)$ it gives $F^* n^2 = 19.80$ against the exact $\ln |I_0| = 22.97$, a factor of 24 too small. The subleading terms come from the same saddle. The discrete maximum behaves as $L_{\max} = F^* n^2 + 3n \ln n + 3n \int g \ln(2\pi g) + \frac{1}{4} \ln n + O(1)$, where g is the density of one arc: the $n \ln n$ and $O(n)$ terms are the self-energy of each species at local spacing $1/(ng)$, which we verified to six digits, and the $\frac{1}{4}$ is consistent with $\frac{1}{24}$ per soft edge. Adding the Gaussian fluctuations about the maximum overshoots by a universal amount, the excess of the harmonic approximation for a $\beta = 2$ log gas, which on the circle is $\ln(e/\sqrt{2\pi})$ per eigenvalue up to $O(1/n)$. With that correction the estimate reproduces (9) to $1/36n$, and

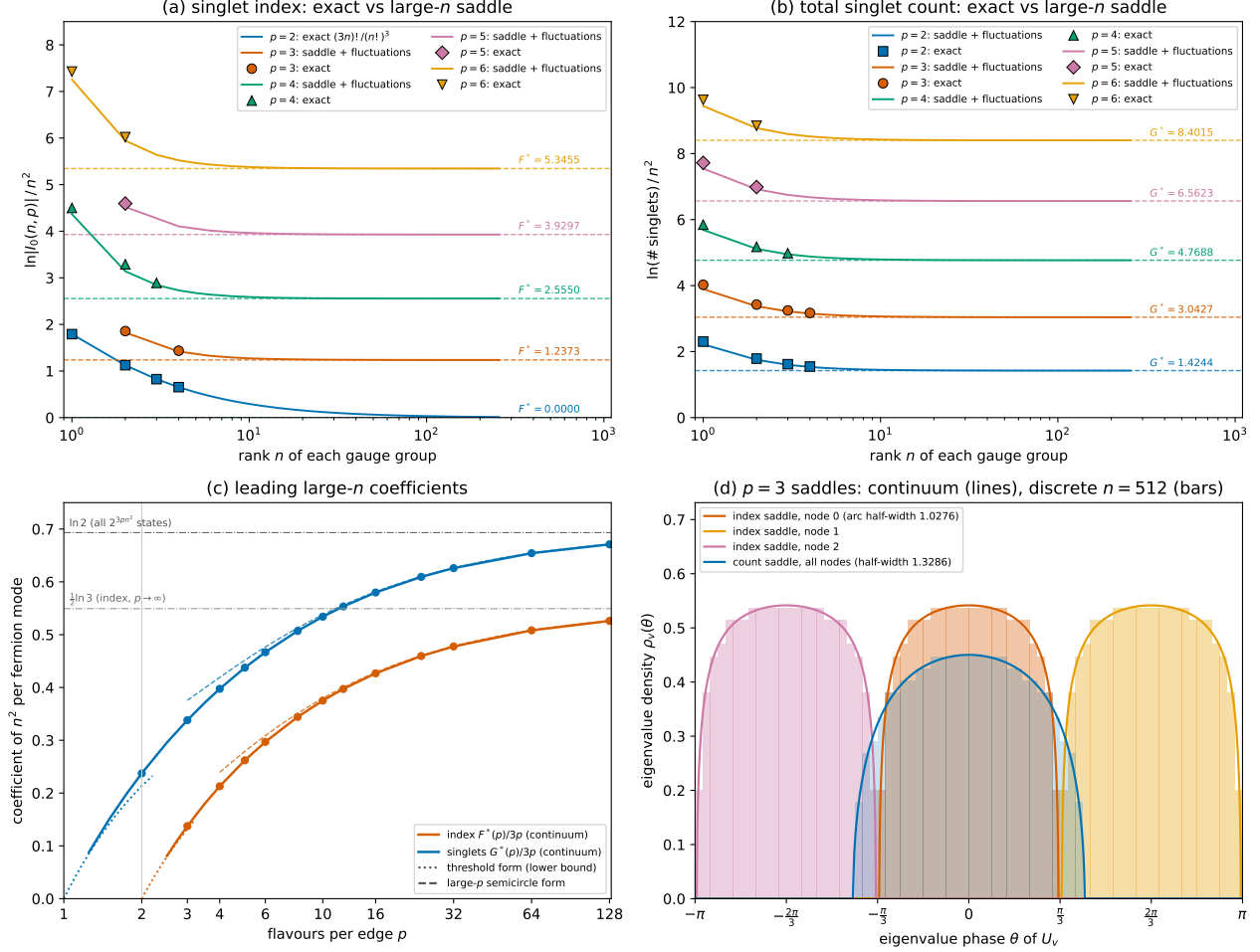


Figure 4: The large- n saddle. (a) $\ln|I_0|/n^2$ against n : exact values (markers), the saddle-point estimate with Gaussian fluctuations and the universal crystal correction (lines), and the asymptotes $F^*(p)$ (dashed); for $p=2$ the line is the exact result (9). (b) The same for the total singlet count. (c) The leading coefficients per fermion mode against p , with the threshold bound and the large- p asymptotics. (d) The equilibrium densities at $p=3$: the three index arcs and the single arc of the count saddle, with histograms of the discrete maximum at $n=512$.

for $p \neq 2$ we assume that each arc is locally a $\beta=2$ gas, which is the one heuristic step. The resulting estimate has no free parameters, and Table 3 compares it with every non-zero exact value we have. All 27 residuals are negative and none exceeds 0.6 in the logarithm, across values from $e^{1.8}$ to $e^{50.7}$; at $(4, 3)$ the estimate gives 22.69, within 25% of $|I_0| = 9.43 \times 10^9$. At large n the $O(n)$ terms cancel to fit precision and the estimate takes the form $\ln|I_0| \approx F^*n^2 + \frac{1}{2}\ln n + d(p)$, with $d \approx 2.2$ – 2.4 for $p=3$ – 6 , in line with the absence of an $O(n)$ term in $\beta=2$ ensembles. For $p=3$ it predicts $\ln|I_0| \approx 47.7, 82.6, 127.3$ and 181.8 at $n=6, 8, 10$ and 12 , beyond the reach of the exact recursion.

The status of these statements should be clear. The rewriting (8), the formula (9) and the sign rules are exact. The coefficient F^* for even p rests on standard large-deviation results for log gases, which we have not reproved. For odd p it also requires that the sign σ^p does not cancel the leading saddle. Sign changes need an eigenvalue of one node to pass an eigenvalue of a neighbouring node across the gap between arcs, which costs an action of order n ; but the gap at $p=3$ is only 0.039 radians, so the suppression is modest at the sizes of interest. The exact values at $(2, 3)$ and $(4, 3)$ show no sign of cancellation, their residuals being like those at even p . Finally, $|I_0|$ is a lower bound on the number of singlet BPS states, equal to it only in the absence of cancellations between degrees, for example under concentration; and for odd pn^2 the index vanishes and says nothing about that number.

(n, p)	$\ln I_0 $			$\ln(\#\text{singlets})$		
	exact	saddle	difference	exact	saddle	difference
(1, 2)	1.79	1.76	-0.03	2.30	2.22	-0.08
(2, 2)	4.50	4.49	-0.01	7.15	7.00	-0.15
(3, 2)	7.43	7.42	-0.01	14.56	14.40	-0.17
(4, 2)	10.45	10.45	-0.01	24.75	24.55	-0.19
(1, 3)		both vanish		4.03	3.90	-0.13
(2, 3)	7.43	7.30	-0.12	13.70	13.50	-0.20
(3, 3)		both vanish		29.22	28.97	-0.25
(4, 3)	22.97	22.69	-0.28	50.74	50.44	-0.30
(1, 4)	4.50	4.37	-0.13	5.85	5.69	-0.16
(2, 4)	13.16	12.56	-0.60	20.69	20.45	-0.24
(3, 4)	26.02	25.62	-0.39	44.85	44.54	-0.30
(1, 5)		both vanish		7.72	7.54	-0.18
(2, 5)	18.38	18.08	-0.29	27.94	27.68	-0.26
(1, 6)	7.43	7.26	-0.17	9.63	9.44	-0.19
(2, 6)	24.08	23.78	-0.30	35.37	35.09	-0.28

Table 3: The saddle-point estimate (Gaussian fluctuations about the discrete saddle, minus the universal $\beta = 2$ crystal excess, no free parameters) against all exact values. For odd p and odd n the index and the estimate both vanish exactly, the latter because the two orientations of the arcs cancel.

property	$p = 2$	$p = 3$ (and $p \geq 3$)
BPS entropy	$\ln I_0 = 3n \ln 3 + O(\ln n)$, proved; not macroscopic	$\ln I_0 = F^*(p)n^2 + O(\ln n)$, $F^*(3) = 1.2373$, 41% of the singlet log-count (large-deviation level; sign proviso for odd p)
Concentration	(2, 2): all 90 singlet BPS states at $k = 12$, exact	(2, 3): all 1680 at $k = 18$, numerical; $n \geq 3$ open; index signs consistent
Fortuity	(2, 2): all 90 classes obstructed at first order	untested; follows from concentration at the next rank
Non-BPS statistics	not random-matrix; hidden $\mathcal{N} = 4$ on singlets	Gaussian orthogonal, 10 024 levels at (2, 3)
Near-BPS gap	none (≥ 95.6)	0.054 next to the BPS degree; Turiaci–Witten not yet testable
Supersymmetry	$\mathcal{N} = 4$ on singlets (numerical, $n = 2$)	$\mathcal{N} = 2$
Large n	planar; exact index	planar; index saddle solved

Table 4: Desirable properties of the quiver for $p = 2$ and $p \geq 3$, with their status.

7 Prospects for the $p \geq 3$ quiver

Table 4 sets the two cases side by side. The contrast is sharp. The $p = 2$ quiver is an elegant and highly structured corner, with a hidden $\mathcal{N} = 4$ on singlets, an exact index and a spectrum that is not random-matrix like, but its BPS entropy is provably sub-macroscopic. The $p \geq 3$ quiver has every property we have been able to test at the level we would hope for: a singlet BPS entropy that is a finite fraction of the total at order n^2 , a concentrated BPS sector, chaotic non-BPS spectra, small gaps next to the BPS degree, and only $\mathcal{N} = 2$ supersymmetry, which is what the $\mathcal{N} = 2$ super-Schwarzian needs. The large- p limit adds a pleasing interpolation: the singlet BPS entropy per fermion rises from zero at $p = 2$ towards the SYK value at $p = \infty$, so the flavour number dials between a gauge-dominated matrix regime and an SYK-like one, with planar large- n control at every p .

The caveats are equally clear, and they define the programme. Almost everything except the entropy has been established at $n = 2$ only, and concentration is the property most at risk as n grows: in the adjoint models it was the growth of the window with the rank that failed, and although its mechanism is absent here, nothing yet proves that another one does not take its place. The near-BPS gaps at (2, 3) are suggestive, but at this size the neighbouring charge sectors differ too much in entropy for the Turiaci–Witten density to apply, so the super-Schwarzian question remains open. Fortuity at $p \geq 3$ is untested, although it would follow from concentration at the next rank and there is no reason to expect a different answer from (2, 2). And the model is planar rather than melonic, so at fixed p there is no analogue of the SYK Schwinger–Dyson solution; large- n dynamics has to

come from the bootstrap, from other matrix-model methods, or from the large- p limit discussed below.

Several next steps follow, and the first observation is that concentration beyond $n = 2$ has to be settled without help from the index. Since Q raises the fermion number by three and every singlet has $k \equiv 0 \pmod 3$, no R-charge fugacity is compatible with the pairing of non-BPS states in the singlet sector: $\text{Tr}(-1)^k y^{k/3}$ is protected only at $y = 1$. The singlet index is therefore the only protected quantity of the gauged model, and its sign, which always agrees with half filling, is all it can say about where the BPS states sit.

The exact methods reach exactly one more point. With a singlet-adapted basis, built from the dual-Cauchy components with Q acting as Pieri maps, the singlet spaces at $(3, 2)$ have 3 494 states at $k = 12$ and 558 304 at half filling, so concentration, the near-BPS gap and the multiplet statistics can be followed from $n = 2$ to $n = 3$, albeit in the $p = 2$ corner. The same basis at $(2, 3)$, 300 584 singlets at half filling, yields the 1680 BPS vectors for a direct test of BPS chaos and of fortuity at $p = 3$. Beyond that the singlet sectors explode: at $(3, 3)$ they hold 1.2×10^8 states at $k = 18$ and 1.1×10^{12} at half filling, and 3×10^8 at half filling at $(2, 4)$.

At $p = 3$ and $n \geq 3$ the matrix bootstrap [15, 16, 17], which we developed in sector-resolved form for the adjoint models [7], must take over, and its natural first targets are certificates of absence. A positive lower bound on the singlet energy at $(3, 3)$ and $k = 18$, far from the window where such certificates are easiest, would prove that all 1680 classes at $(2, 3)$ are fortuitous; certificates for $k \leq 36$ would confine the BPS states at $(3, 3)$ to the two singlet degrees adjacent to half filling, which is concentration at the first rank beyond exact methods. The near-BPS gaps and their scaling with n , the actual super-Schwarzian test, are the harder final target.

On the analytic side, the large- p limit is where the model should become SYK-like. The holonomy saddle realises precisely the $\mathbb{Z}_3$ twist that makes the index of $\mathcal{N} = 2$ SYK large, and at fixed n the $3pn^2$ fermions with random flavour couplings should be governed by the Schwinger–Dyson equations of [3], with a gauge correction of relative order $1/p$ since $\dim G / \dim V = 1/p$. Deriving the super-Schwarzian and the chaos exponent in this limit, and reproducing the exact large- p expansion of $F^*(p)$ in Table 2 as a check, would establish the desired dynamics in a controlled corner of the same model; that $F^*(p)$ is smooth for all $p > 2$ suggests that $p = 3$ lies in the same phase. Monte Carlo integration of (8), by thermodynamic integration in p from the exact point $p = 2$, would test the finite- n predictions at $n = 6$ –12, and the generalised monotones of [14] should be checked. Finally, the hidden $\mathcal{N} = 4$ of the $p = 2$ quiver deserves a derivation and a test at $n = 3$, both for its own sake and because a small $\mathcal{N} = 4$ super-Schwarzian would require an $SU(2)$ R-symmetry that we have not found.

In summary, the fermionic $U(n)^3$ quiver with $p \geq 3$ flavours is, to our knowledge, a new candidate for a matrix SYK model. It has macroscopic and, at the sizes we can reach, concentrated BPS entropy, chaotic multiplet spectra and near-BPS gaps; it reduces to an exactly solvable log gas at the level of the index; and its large- p limit makes contact with $\mathcal{N} = 2$ SYK. Whether concentration survives at large n is now the central question; the singlet-adapted computation at $(3, 2)$, bootstrap certificates at $(3, 3)$ and the large- p limit are the next steps.

A Methods and reproducibility

All computations are reproducible from the repository, with the parameters, seeds, outputs and memory use recorded in the result files. Cohomology at $(2, 2)$ uses exact ranks modulo primes. The $n = 2$ spectra use Weyl-reduced zero-weight sectors (`src/quiver_n2.py`) with the exact Casimir penalty, Lanczos (`scripts/quiver_singlet_zero_modes.py`), dense singlet bases (`scripts/quiver_singlet_spectrum.py`), the Casimir-polynomial projection for the 10 024-level pair (`scripts/quiver_singlet_pair_large.py`, float64, peak 8.6 GB) and the $\mathcal{N} = 4$ and symmetry analysis (`scripts/quiver_p2_n4.py`, `scripts/quiver_commutant.py`). The fortuity test is `scripts/run_quiver_fortuity.py`. Singlet indices use `src/quiver_index.py` (dual Cauchy and the Littlewood–Richardson transfer matrix, exact integers). The log-gas analysis is `src/quiver_saddle.py` and `scripts/quiver_index_saddle.py`, tested by `tests/test_quiver_saddle.py` (reduced against full energies, Hessians against finite differences to 10^{-10} , the $p = 2$ formula, the continuum values, and the eigenvalue representation against direct quadrature at $n = 1$). The derivations are written out in `docs/derivations.md` (D18) and the research log in `research/notes/quiver_project.md`. The main data files are `results/data/quiver_singlet_index_dixon.json`, `quiver_singlet_zero_modes_n2_p3_k*.json`, `quiver_singlet_pair_n2_p3_k9_f64.json`, `quiver_p2_n4_*.json`, `quiver_fortuity_n2_p2_k12.json` and `quiver_index_saddle.json`.

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